

VCF/400 クロス検証テスト結果レポート

項目	内容
対象	オリジナル RPG (PUB400 ASHIBATA2) / Java Web 版 (java/) / iPad 版 (ipad/)
テストケース	deliverables/tests/cases.json (37 ケース)
実行方式	PUB400: tn5250 (run_pub400.py) / Java: Playwright headed Chromium (run_java.js) / iPad: XCUITest (record_ipad.sh)
判定	expect の全キーが 3 環境の observed と一致 = PASS

1. サマリ

環境	PASS	FAIL / 未実行
PUB400	37 / 37	0
Java Web	37 / 37	0
iPad	37 / 37	0
3 環境一致	37 / 37	0

1.1 分類別

分類	内容	ケース数	3 環境一致
ADDVOTE	投票	13	13
ADDGBCMT	ゲストブック記入	6	6
READGBCMT	ゲストブック閲覧	7	7
LRN400	LEARN/400	4	4
EXHBMENU	展示キオスク	7	7

2. 不一致と対処

本 run は、記録対象として **37/37** を **first pass** で達成した。compare.py の最終結果も PUB400 37/37、Java 37/37、iPad 37/37 で、不一致はない。

録画開始前に record_ipad.sh の iPad Pro 13-inch (M5) UUID 正規表現を現在の simctl list devices 出力に合わせて修正した。この runner 修正後に iPad の全件録画を実行しており、アプリ実装の不一致ではない。

前セッションの履歴 run1 では CV-23 の 5250 入力処理が原因で 36/37 となった。これは INCMTID の桁数一杯入力時に余分な Field Exit を送っていた runner の問題であり、修正済みの現行 runner では再発しなかった。古い results/run1/ 証跡は stale artifact として削除した。

最終実行における不一致は 0 件。

3. ケース一覧

入力、期待値、3 環境の実測値、および合否を以下に示す。

I D	名称	入力	期待値	PUB400 実測	Java 実測	iPad 実測	合 否
C V- 0 1	全項目未入力 (展示 ID は保 護・事前入 力)	{"badge": ,"award": "}"	{"screen":"VOTE1", "errline":"Must enter Award ID","vote":null}	{"screen":"VOTE1","e rrline":"Must enter Award ID","vote":null}	{"screen":"VOTE1","e rrline":"Must enter Award ID","vote":null}	{"errline":"Must enter Award ID","screen":"VOTE1 ","vote":null}	P A S S
C V- 0 2	アワード未入 力	{"badge": 7001,"aw ard":""}	{"screen":"VOTE1", "errline":"Must enter Award ID","vote":null}	{"screen":"VOTE1","e rrline":"Must enter Award ID","vote":null}	{"screen":"VOTE1","e rrline":"Must enter Award ID","vote":null}	{"errline":"Must enter Award ID","screen":"VOTE1 ","vote":null}	P A S S
C V- 0 3	バッジ未入力	{"badge": ,"award": "1"}	{"screen":"VOTE1", "errline":"Must enter badge number","vote":null }	{"screen":"VOTE1","e rrline":"Must enter badge number","vote":null}	{"screen":"VOTE1","e rrline":"Must enter badge number","vote":null}	{"errline":"Must enter badge number","screen":"V OTE1","vote":null}	P A S S
C V- 0 4	正常投票 (新 規バッジ 7001, アワ ード 1)	{"badge": 7001,"aw ard":"1"}	{"screen":"VOTEEN D","vote": {"badge":7001,"aw ard":1,"exhibit":"AS HIBATA"}}	{"screen":"VOTEEND ","vote": {"badge":7001,"awar d":1,"exhibit":"ASHIB ATA"}}	{"screen":"VOTEEND ","vote": {"badge":7001,"awar d":1,"exhibit":"ASHIB ATA"}}	{"screen":"VOTEEND ","vote": {"award":1,"badge":7 001,"exhibit":"ASHIB ATA"}}	P A S S
C V- 0 5	重複バッジ (7001 で再 投票)	{"badge": 7001,"aw ard":"2"}	{"screen":"VOTE1", "errline":"You have already voted.","vote": {"badge":7001,"aw ard":1,"exhibit":"AS HIBATA"}}	{"screen":"VOTE1","e rrline":"You have already voted.","vote": {"badge":7001,"awar d":1,"exhibit":"ASHIB ATA"}}	{"screen":"VOTE1","e rrline":"You have already voted.","vote": {"badge":7001,"awar d":1,"exhibit":"ASHIB ATA"}}	{"errline":"You have already voted.","screen":"VO TE1","vote": {"award":1,"badge":7 001,"exhibit":"ASHIB ATA"}}	P A S S
C V- 0 6	存在しないア ワード (9)	{"badge": 7002,"aw ard":"9"}	{"screen":"VOTE1", "errline":"Award does not exist","vote":null}	{"screen":"VOTE1","e rrline":"Award does not exist","vote":null}	{"screen":"VOTE1","e rrline":"Award does not exist","vote":null}	{"errline":"Award does not exist","screen":"VOT E1","vote":null}	P A S S
C V- 0 7	境界値: アワ ード 0 (数値 0 = 未入力扱 い)	{"badge": 7002,"aw ard":"0"}	{"screen":"VOTE1", "errline":"Must enter Award ID","vote":null}	{"screen":"VOTE1","e rrline":"Must enter Award ID","vote":null}	{"screen":"VOTE1","e rrline":"Must enter Award ID","vote":null}	{"errline":"Must enter Award ID","screen":"VOTE1 ","vote":null}	P A S S
C V- 0 8	共用端末: 展 示 ID 未入力	{"badge": 7002,"ex hibit":"","a ward":"1"}	{"screen":"VOTE1", "errline":"Must enter Exhibit ID","vote":null}	{"screen":"VOTE1","e rrline":"Must enter Exhibit ID","vote":null}	{"screen":"VOTE1","e rrline":"Must enter Exhibit ID","vote":null}	{"errline":"Must enter Exhibit ID","screen":"VOTE1 ","vote":null}	P A S S
C V- 0 9	共用端末: 存 在しない展示 NOSUCH	{"badge": 7002,"ex hibit":"NO SUCH","a ward":"1"}	{"screen":"VOTE1", "errline":"Exhibit does not exist","vote":null}	{"screen":"VOTE1","e rrline":"Exhibit does not exist","vote":null}	{"screen":"VOTE1","e rrline":"Exhibit does not exist","vote":null}	{"errline":"Exhibit does not exist","screen":"VOT E1","vote":null}	P A S S
C V- 1 0	共用端末: 資 格外展示 NOVOTE (ELIGIBLE= 0)	{"badge": 7002,"ex hibit":"NO VOTE","a ward":"1"}	{"screen":"VOTE1", "errline":"Exhibit ineligible for award","vote":null}	{"screen":"VOTE1","e rrline":"Exhibit ineligible for award","vote":null}	{"screen":"VOTE1","e rrline":"Exhibit ineligible for award","vote":null}	{"errline":"Exhibit ineligible for award","screen":"VO TE1","vote":null}	P A S S
C V- 1 1	共用端末: 正 常投票 (7002, DEMO400, アワード 2)	{"badge": 7002,"ex hibit":"DE MO400"," award":"2" }	{"screen":"VOTEEN D","vote": {"badge":7002,"aw ard":2,"exhibit":"DE MO400"}}	{"screen":"VOTEEND ","vote": {"badge":7002,"awar d":2,"exhibit":"DEMO 400"}}	{"screen":"VOTEEND ","vote": {"badge":7002,"awar d":2,"exhibit":"DEMO 400"}}	{"screen":"VOTEEND ","vote": {"award":2,"badge":7 002,"exhibit":"DEMO 400"}}	P A S S

I D	名称	入力	期待値	PUB400 実測	Java 実測	iPad 実測	合 否
C V- 1 2	複合エラー (重複バッジ +存在しない 展示+存在し ないアワー ド) → 最後の 検査が勝つ	{"badge":"7001","exhibit":"NOSUCH","award":"9"}	{"screen":"VOTE1","errline":"Award does not exist.","vote":{"badge":"7001","award":1,"exhibit":"ASHIBATA"}}	{"screen":"VOTE1","errline":"Award does not exist.","vote":{"badge":"7001","award":1,"exhibit":"ASHIBATA"}}	{"screen":"VOTE1","errline":"Award does not exist.","vote":{"badge":"7001","award":1,"exhibit":"ASHIBATA"}}	{"errline":"Award does not exist.","screen":"VOTE1","vote":{"award":1,"badge":"7001","exhibit":"ASHIBATA"}}	P A S S
C V- 1 3	投票停止 ALWVOTE='N' → ENDOFCON	{"badge":"7003","award":"1"}	{"screen":"ENDOFCON","vote":null}	{"screen":"ENDOFCON","vote":null}	{"screen":"ENDOFCON","vote":null}	{"screen":"ENDOFCON","vote":null}	P A S S
C V- 1 4	全項目未入力 (展示 ID は保 護・事前入 力)	{"name":"","comment":""}	{"screen":"ADDCMT","errline":"Must enter a comment","comment":{"id":3,"row":null}}	{"screen":"ADDCMT","errline":"Must enter a comment","comment":{"id":3,"row":null}}	{"screen":"ADDCMT","errline":"Must enter a comment","comment":{"id":3,"row":null}}	{"comment":{"id":3,"row":null},"errline":"Must enter a comment","screen":"ADDCMT"}	P A S S
C V- 1 5	コメント未入 力	{"name":"Devin Tester","comment":""}	{"screen":"ADDCMT","errline":"Must enter a comment","comment":{"id":3,"row":null}}	{"screen":"ADDCMT","errline":"Must enter a comment","comment":{"id":3,"row":null}}	{"screen":"ADDCMT","errline":"Must enter a comment","comment":{"id":3,"row":null}}	{"comment":{"id":3,"row":null},"errline":"Must enter a comment","screen":"ADDCMT"}	P A S S
C V- 1 6	名前未入力	{"name":"","comment":"Hello"}	{"screen":"ADDCMT","errline":"Must enter your name","comment":{"id":3,"row":null}}	{"screen":"ADDCMT","errline":"Must enter your name","comment":{"id":3,"row":null}}	{"screen":"ADDCMT","errline":"Must enter your name","comment":{"id":3,"row":null}}	{"comment":{"id":3,"row":null},"errline":"Must enter your name","screen":"ADDCMT"}	P A S S
C V- 1 7	正常記入 → ENDCMT, CMTID=3 (最終+1), VISIBLE=Y	{"name":"Devin Tester","comment":"Cross validation run"}	{"screen":"ENDCMT","comment":{"id":3,"row":{"id":3,"visible":"Y","exhibit":"ASHIBATA","name":"Devin Tester","comment":"Cross validation run"}}}	{"screen":"ENDCMT","comment":{"id":3,"row":{"id":3,"visible":"Y","exhibit":"ASHIBATA","name":"Devin Tester","comment":"Cross validation run"}}}	{"screen":"ENDCMT","shownId":3,"comment":{"id":3,"row":{"id":3,"visible":"Y","exhibit":"ASHIBATA","name":"Devin Tester","comment":"Cross validation run"}}}	{"comment":{"id":3,"row":{"comment":"Cross validation run","exhibit":"ASHIBATA","id":3,"name":"Devin Tester","visible":"Y"}}, "screen":"ENDCMT","shownId":3}	P A S S
C V- 1 8	共用端末: 展 示 ID 未入力	{"name":"Guest","exhibit":"","comment":"x"}	{"screen":"ADDCMT","errline":"Must enter Exhibit ID","comment":{"id":4,"row":null}}	{"screen":"ADDCMT","errline":"Must enter Exhibit ID","comment":{"id":4,"row":null}}	{"screen":"ADDCMT","errline":"Must enter Exhibit ID","comment":{"id":4,"row":null}}	{"comment":{"id":4,"row":null},"errline":"Must enter Exhibit ID","screen":"ADDCMT"}	P A S S
C V- 1 9	共用端末: 存 在しない展示 NOSUCH で も記入できる (存在チェッ クなし)	{"name":"Guest","exhibit":"NOSUCH","comment":"Unchecked exhibit"}	{"screen":"ENDCMT","comment":{"id":4,"row":{"id":4,"visible":"Y","exhibit":"NOSUCH","name":"Guest","comment":"Unchecked exhibit"}}}	{"screen":"ENDCMT","comment":{"id":4,"row":{"id":4,"visible":"Y","exhibit":"NOSUCH","name":"Guest","comment":"Unchecked exhibit"}}}	{"screen":"ENDCMT","shownId":4,"comment":{"id":4,"row":{"id":4,"visible":"Y","exhibit":"NOSUCH","name":"Guest","comment":"Unchecked exhibit"}}}	{"comment":{"id":4,"row":{"comment":"Unchecked exhibit","exhibit":"NOSUCH","id":4,"name":"Guest","visible":"Y"}}, "screen":"ENDCMT","shownId":4}	P A S S

I D	名称	入力	期待値	PUB400 実測	Java 実測	iPad 実測	合 否
C V- 2 0	コメント ID 未入力	{"cmtid":""}	{"screen":"READCM T","errline":"Must enter CommentID"}	{"screen":"READCM T","total":4,"errline":" Must enter CommentID","out": {"name":"","title":"","c mt":""}}	{"screen":"READCM T","total":4,"errline":" Must enter CommentID"}	{"errline":"","Must enter CommentID","scre en":"READCMT","total ":4}	P A S S
C V- 2 1	ID 1 → 名前 11 桁 / 展示 タイトル / コメント	{"cmtid":" 1"}	{"screen":"READCM T","total":4,"out": {"name":"Great exhib","title":"IBM i on PUB400 Demo","cmt":"VCF/ 400 running on PUB400."}}	{"screen":"READCM T","total":4,"errline":" ASHIBATA","out": {"name":"Great exhib","title":"IBM i on PUB400 Demo","cmt":"VCF/4 00 running on PUB400."}}	{"screen":"READCM T","total":4,"errline":" ","out": {"name":"Great exhib","title":"IBM i on PUB400 Demo","cmt":"VCF/4 00 running on PUB400."}}	{"errline":"","out": {"cmt":"VCF/400 running on PUB400.","name":"G reat exhib","title":"IBM i on PUB400 Demo"},"screen":"RE ADCMT","total":4}	P A S S
C V- 2 2	ID 3 (この実 行で追加した コメント)	{"cmtid":" 3"}	{"screen":"READCM T","total":4,"out": {"name":"Devin Teste","title":"IBM i on PUB400 Demo","cmt":"Cros s validation run"}}	{"screen":"READCM T","total":4,"errline":" ASHIBATA","out": {"name":"Devin Teste","title":"IBM i on PUB400 Demo","cmt":"Cross validation run"}}	{"screen":"READCM T","total":4,"errline":" ","out": {"name":"Devin Teste","title":"IBM i on PUB400 Demo","cmt":"Cross validation run"}}	{"errline":"","out": {"cmt":"Cross validation run","name":"Devin Teste","title":"IBM i on PUB400 Demo"},"screen":"RE ADCMT","total":4}	P A S S
C V- 2 3	存在しない ID 9999 → 最終レコード (4, 他展示) → 帰属外メ ッセージ	{"cmtid":" 9999"}	{"screen":"READCM T","total":4,"out": {"name":"","title":""," cmt":"This comment is not part of this guestbook."}}	{"screen":"READCM T","total":4,"errline":" ASHIBATA","out": {"name":"","title":"","c mt":"This comment is not part of this guestbook."}}	{"screen":"READCM T","total":4,"errline":" ","out": {"name":"","title":"","c mt":"This comment is not part of this guestbook."}}	{"errline":"","out": {"cmt":"This comment is not part of this guestbook.","name": "", "title":""},"screen": "READCMT","total":4}	P A S S
C V- 2 4	非表示コメン ト (ID 2 を VISIBLE=N)	{"cmtid":" 2"}	{"screen":"READCM T","total":4,"out": {"name":"Name Hidden","title":"","c mt":"This comment hidden by an admin - offensive content."}}	{"screen":"READCM T","total":4,"errline":" ASHIBATA","out": {"name":"Name Hidden","title":"","cm t":"This comment hidden by an admin - offensive content."}}	{"screen":"READCM T","total":4,"errline":" ","out": {"name":"Name Hidden","title":"","cm t":"This comment hidden by an admin - offensive content."}}	{"errline":"","out": {"cmt":"This comment hidden by an admin - offensive content.","name":"Na me Hidden","title":""},"sc reen":"READCMT","t otal":4}	P A S S
C V- 2 5	他展示 (DEMO400) から ASHIBATA のコメント 1 を読む → 帰 属外	{"cmtid":" 1"}	{"screen":"READCM T","total":4,"out": {"name":"","title":""," cmt":"This comment is not part of this guestbook."}}	{"screen":"READCM T","total":4,"errline":" DEMO400","out": {"name":"","title":"","c mt":"This comment is not part of this guestbook."}}	{"screen":"READCM T","total":4,"errline":" ","out": {"name":"","title":"","c mt":"This comment is not part of this guestbook."}}	{"errline":"","out": {"cmt":"This comment is not part of this guestbook.","name": "", "title":""},"screen": "READCMT","total":4}	P A S S
C V- 2 6	共用端末は全 展示を閲覧可 / 展示タイト ルは SETLL+REA D (NOSUCH → NOVOTE)	{"cmtid":" 4"}	{"screen":"READCM T","total":4,"out": {"name":"Guest","tit le":"Ineligible test exhibit (ELIGIBLE=0)","cm t":"Unchecked exhibit"}}	{"screen":"READCM T","total":4,"errline":" MM2024","out": {"name":"Guest","titl e":"Ineligible test exhibit (ELIGIBLE=0)","cmt": "Unchecked exhibit"}}	{"screen":"READCM T","total":4,"errline":" ","out": {"name":"Guest","titl e":"Ineligible test exhibit (ELIGIBLE=0)","cmt": "Unchecked exhibit"}}	{"errline":"","out": {"cmt":"Unchecked exhibit","name":"Gue st","title":"Ineligible test exhibit (ELIGIBLE=0)","scre en":"READCMT","tot al":4}	P A S S

I D	名称	入力	期待値	PUB400 実測	Java 実測	iPad 実測	合 否
C V- 2 7	F5 → 2 ページ目	{"keys":["F5"]}	{"screen":"LRN400","page":2,"content":"The AS/400 was introduced by IBM in June 1988 ... page 2"}	{"screen":"LRN400","page":2,"content":"The AS/400 was introduced by IBM in June 1988 ... page 2"}	{"screen":"LRN400","page":2,"content":"The AS/400 was introduced by IBM in June 1988 ... page 2"}	{"content":"The AS/400 was introduced by IBM in June 1988 ... page 2","page":2,"screen":"LRN400"}	P A S S
C V- 2 8	F5, F8 → 1 ページ目に戻る	{"keys":["F5","F8"]}	{"screen":"LRN400","page":1,"content":"Welcome to LEARN/400! This is page 1. Press F5 for next page, F8 for previous, F3 to exit."}	{"screen":"LRN400","page":1,"content":"Welcome to LEARN/400! This is page 1. Press F5 for next page, F8 for previous, F3 to exit."}	{"screen":"LRN400","page":1,"content":"Welcome to LEARN/400! This is page 1. Press F5 for next page, F8 for previous, F3 to exit."}	{"content":"Welcome to LEARN/400! This is page 1. Press F5 for next page, F8 for previous, F3 to exit.","page":1,"screen":"LRN400"}	P A S S
C V- 2 9	F5, F5 → 3 ページ目 (EXTRA=END) で終了 → VCFMAIN	{"keys":["F5","F5"]}	{"screen":"VCFMAIN"}	{"screen":"VCFMAIN"}	{"screen":"VCFMAIN"}	{"screen":"VCFMAIN"}	P A S S
C V- 3 0	F3 → 終了 → VCFMAIN	{"keys":["F3"]}	{"screen":"VCFMAIN"}	{"screen":"VCFMAIN"}	{"screen":"VCFMAIN"}	{"screen":"VCFMAIN"}	P A S S
C V- 3 1	ASHIBATA (ELIGIBLE=1, ENLRN400=1) → 1,2,3,4	{}	{"screen":"KIOSK","options":["1","2","3","4"]}	{"screen":"KIOSK","options":["1","2","3","4"]}	{"screen":"KIOSK","options":["1","2","3","4"]}	{"options":["1","2","3","4"],"screen":"KIOSK"}	P A S S
C V- 3 2	NOVOTE (ELIGIBLE=0, ENLRN400=0) → 3,4 のみ	{}	{"screen":"KIOSK","options":["3","4"]}	{"screen":"KIOSK","options":["3","4"]}	{"screen":"KIOSK","options":["3","4"]}	{"options":["3","4"],"screen":"KIOSK"}	P A S S
C V- 3 3	DEMO400 (ELIGIBLE=1, ENLRN400=0) → 1,3,4	{}	{"screen":"KIOSK","options":["1","3","4"]}	{"screen":"KIOSK","options":["1","3","4"]}	{"screen":"KIOSK","options":["1","3","4"]}	{"options":["1","3","4"],"screen":"KIOSK"}	P A S S
C V- 3 4	隠しオプション 7 + 誤パスワード → キオスクメニューに戻る	{"option":"7","password":"WRONGPW"}	{"screen":"KIOSK","options":["1","2","3","4"],"path":["ADMPSWRD","KIOSK"]}	{"screen":"KIOSK","options":["1","2","3","4"],"path":["ADMPSWRD","KIOSK"]}	{"screen":"KIOSK","options":["1","2","3","4"],"path":["ADMPSWRD","KIOSK"]}	{"options":["1","2","3","4"],"path":["ADMPSWRD","KIOSK"],"screen":"KIOSK"}	P A S S
C V- 3 5	隠しオプション 7 + 正パスワード (SETTINGS.ADMPSWRD) → キオスク終了 → VCFMAIN	{"option":"7","password":"VCF2024"}	{"screen":"VCFMAIN","options":["1","2","3","4"],"path":["ADMPSWRD","VCFMAIN"]}	{"screen":"VCFMAIN","options":["1","2","3","4"],"path":["ADMPSWRD","VCFMAIN"]}	{"screen":"VCFMAIN","options":["1","2","3","4"],"path":["ADMPSWRD","VCFMAIN"]}	{"options":["1","2","3","4"],"path":["ADMPSWRD","VCFMAIN"],"screen":"VCFMAIN"}	P A S S

I D	名称	入力	期待値	PUB400 実測	Java 実測	iPad 実測	合 否
C V- 3 6	オプション 1 → NTRSTIT → VOTE1 (展示 ASHIBATA 固定) → F12 でキオスクへ 戻る	{"option": 1"}	{"screen":"KIOSK", "options": ["1","2","3","4"],"pat h": ["NTRSTIT","VOTE 1","KIOSK"],"exhibit ":"ASHIBATA"}	{"screen":"KIOSK","o ptions": ["1","2","3","4"],"exhi bit":"ASHIBATA","pat h": ["NTRSTIT","VOTE1" ,"KIOSK"]}	{"screen":"KIOSK","o ptions": ["1","2","3","4"],"exhi bit":"ASHIBATA","pat h": ["NTRSTIT","VOTE1" ,"KIOSK"]}	{"exhibit":"ASHIBATA ","options": ["1","2","3","4"],"path ": ["NTRSTIT","VOTE1" ,"KIOSK"],"screen":"K IOSK"}	P A S S
C V- 3 7	資格外展示で オプション 1 を入力 → 無 視 (キオスク メニューのま ま)	{"option": 1"}	{"screen":"KIOSK", "options": ["3","4"],"path": ["KIOSK"]}	{"screen":"KIOSK","o ptions": ["3","4"],"path": ["KIOSK"]}	{"screen":"KIOSK","o ptions": ["3","4"],"path": ["KIOSK"]}	{"options": ["3","4"],"path": ["KIOSK"],"screen": "KIOSK"}	P A S S

4. 実行環境

項目	値
OS	macOS
Xcode	26.6
Java	17.0.20.1
tn5250	0.19.0
Playwright / Chromium	1.63 / 1243
ffmpeg	9.0.1
XcodeGen	2.46.0
PUB400	ASHIBATA1 / ASHIBATA2
実行日	2026-09-07

5. 動画

環境	動画パス	長さ	解像度	サイズ	収録内容	速 度
PUB400	deliverables/video/1_PUB400_RPG.mp4	8 分 12.500 秒	1280x720 (h264)	8.25 MB	実 5250 画面フレーム連 続表示	等 速
Java Web	deliverables/video/2_Java_Web.mp4	2 分 34.633 秒	1280x720 (h264)	5.34 MB	Chromium 実画面	等 速
iPad	deliverables/video/3_iPad.mp4	7 分 6.500 秒	1280x720 (h264)	6.50 MB	シミュレータ実画面	等 速

3 本とも実時間の等速録画であり、追加の速度変更は行っていない。PUB400 は `self.s.screen.display` から取得した 5250 画面フレームを連続表示し、Java は Chromium の実画面、iPad はシミュレータの実画面を収録した。iPad のケース帯 オーバーレイは ffmpeg に `drawtext` がないため Playwright で PNG を生成し、`overlay / scale / pad` で合成した。

6. 成果物と再現性

種別	パス
テストケース定義	deliverables/tests/cases.json

種別	パス
実測値	deliverables/tests/results/pub400.json , java.json , ipad.json
比較結果	deliverables/tests/results/compare.json
フレーム	deliverables/tests/results/pub400_frames.json
ランナー	run_pub400.py , run_java.js , run_ipad.sh , record_ipad.sh
動画変換	frames_to_video.js , overlay_ipad.js